

NETFLIX

~PROJECT REPORT



INTRODUCTION

Netflix is one of the world's leading streaming platforms, offering thousands of movies and TV shows across different genres and countries. The objective of this project is to analyze Netflix's content library and identify trends related to content type, genres, ratings, release years, duration, and geographical distribution.





EXPLORATORY DATA ANALYSIS

Objective:

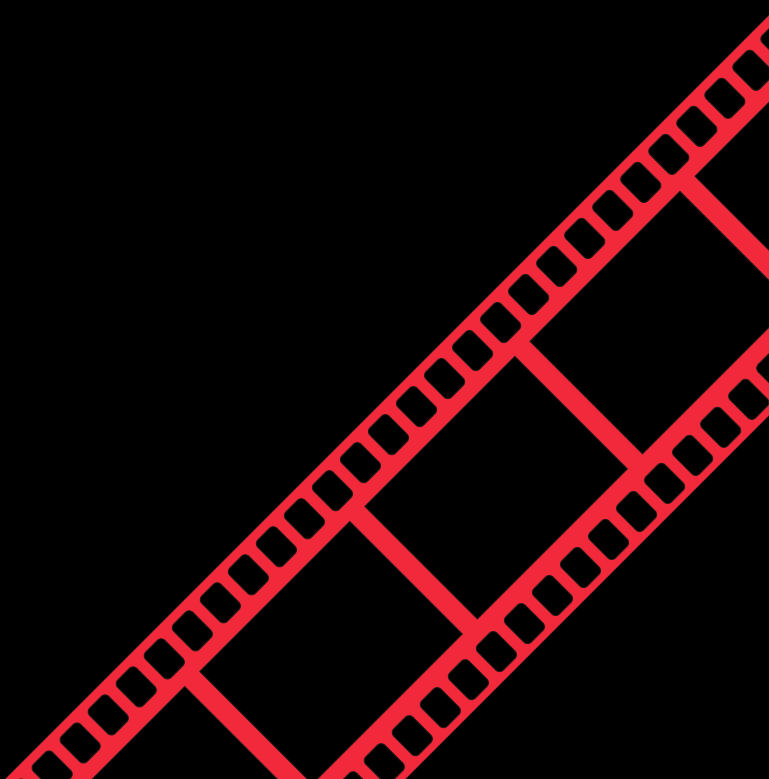
To understand the structure of the dataset and identify important variables available for analysis.

Findings

- Dataset contains information about movies and TV shows available on Netflix.
- Important columns include title, type, country, release year, rating, duration, and genre.
- Both categorical and numerical variables are present.

Insight

The dataset provides sufficient information to understand Netflix's content strategy and growth patterns over time.



EXPLORATORY DATA ANALYSIS

Dataset Overview

The Netflix dataset contains information about movies and TV shows available on the Netflix platform.

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train l...

EXPLORATORY DATA ANALYSIS

Observation

The dataset consists of both categorical and numerical variables, providing information related to content characteristics, geographical origin, and release information.

Insight

The availability of multiple content-related attributes allows comprehensive analysis of Netflix's content library and distribution strategy.



EXPLORATORY DATA ANALYSIS

Dataset Dimensions

Observation

The dataset contains:
8,807 rows (content records)
12 columns (features)

Insight

The dataset size is sufficiently large to identify meaningful trends and patterns in Netflix's content catalog.



EXPLORATORY DATA ANALYSIS

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	show_id	8807 non-null	object
1	type	8807 non-null	object
2	title	8807 non-null	object
3	director	6173 non-null	object
4	cast	7982 non-null	object
5	country	7976 non-null	object
6	date_added	8797 non-null	object
7	release_year	8807 non-null	int64
8	rating	8803 non-null	object
9	duration	8804 non-null	object
10	listed_in	8807 non-null	object
11	description	8807 non-null	object

dtypes: int64(1), object(11)

Data Types Analysis

Observation

The dataset includes:

- Object (Categorical) Variables
- Integer Variables
- Date Variables (after conversion)

Most columns are categorical because they describe content characteristics rather than numerical measurements.

Insight

Categorical features such as genre, country, rating, and content type play a significant role in understanding Netflix's content distribution.

EXPLORATORY DATA ANALYSIS

Missing Values Analysis

Observation

Missing values were identified primarily in:

- Director
- Cast
- Country

These missing values could affect analysis if not handled appropriately.

Insight

Proper treatment of missing values is essential to ensure accurate visualization and statistical analysis.

```
show_id      0
type         0
title        0
director     2634
cast         825
country      831
date_added   10
release_year  0
rating       4
duration     3
listed_in    0
description  0
dtype: int64
```



EXPLORATORY DATA ANALYSIS

Duplicate Records Analysis

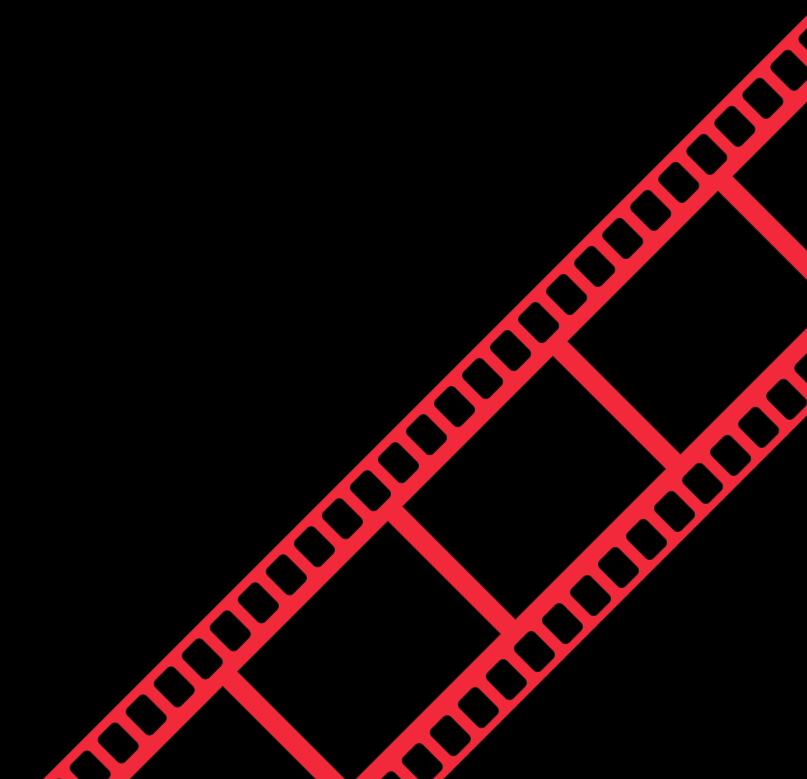
Observation

Duplicate records were checked to identify repeated entries in the dataset.

Insight

Removing duplicate records improves data quality and prevents biased analytical results.

```
df.drop_duplicates(inplace=True)
df['director']=df['director'].fillna('Unknown')
df['cast']=df['cast'].fillna('Unknown')
df['country']=df['country'].fillna('Unknown')
df['date_added']=pd.to_datetime(df['date_added'], errors='coerce')
df['date_added']=df['date_added'].dt.strftime('%Y-%m-%d')
```



EXPLORATORY DATA ANALYSIS

Descriptive Statistics

Objective

To summarize numerical information available in the dataset.

Findings

- Release years span multiple decades.
- Movie durations vary significantly.
- Majority of content belongs to recent years.

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
count	8807	8807	8807	8807	8807	8807	8709	8807.000000	8803	8804	8807	8807
unique	8807	2	8807	4529	7693	749	NaN	NaN	17	220	514	8775
top	s1	Movie	Dick Johnson Is Dead	Unknown	Unknown	United States	NaN	NaN	TV-MA	1 Season	Dramas, International Movies	Paranormal activity at a lush, abandoned prope...
freq	1	6131	1	2634	825	2818	NaN	NaN	3207	1793	362	4
mean	NaN	NaN	NaN	NaN	NaN	NaN	2019-05-23 01:45:29.452290816	2014.180198	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	2008-01-01 00:00:00	1925.000000	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	2018-04-20 00:00:00	2013.000000	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	2019-07-12 00:00:00	2017.000000	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN	NaN	NaN	2020-08-26 00:00:00	2019.000000	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN	NaN	NaN	2021-09-25 00:00:00	2021.000000	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.819312	NaN	NaN	NaN	NaN

VISUALIZATION

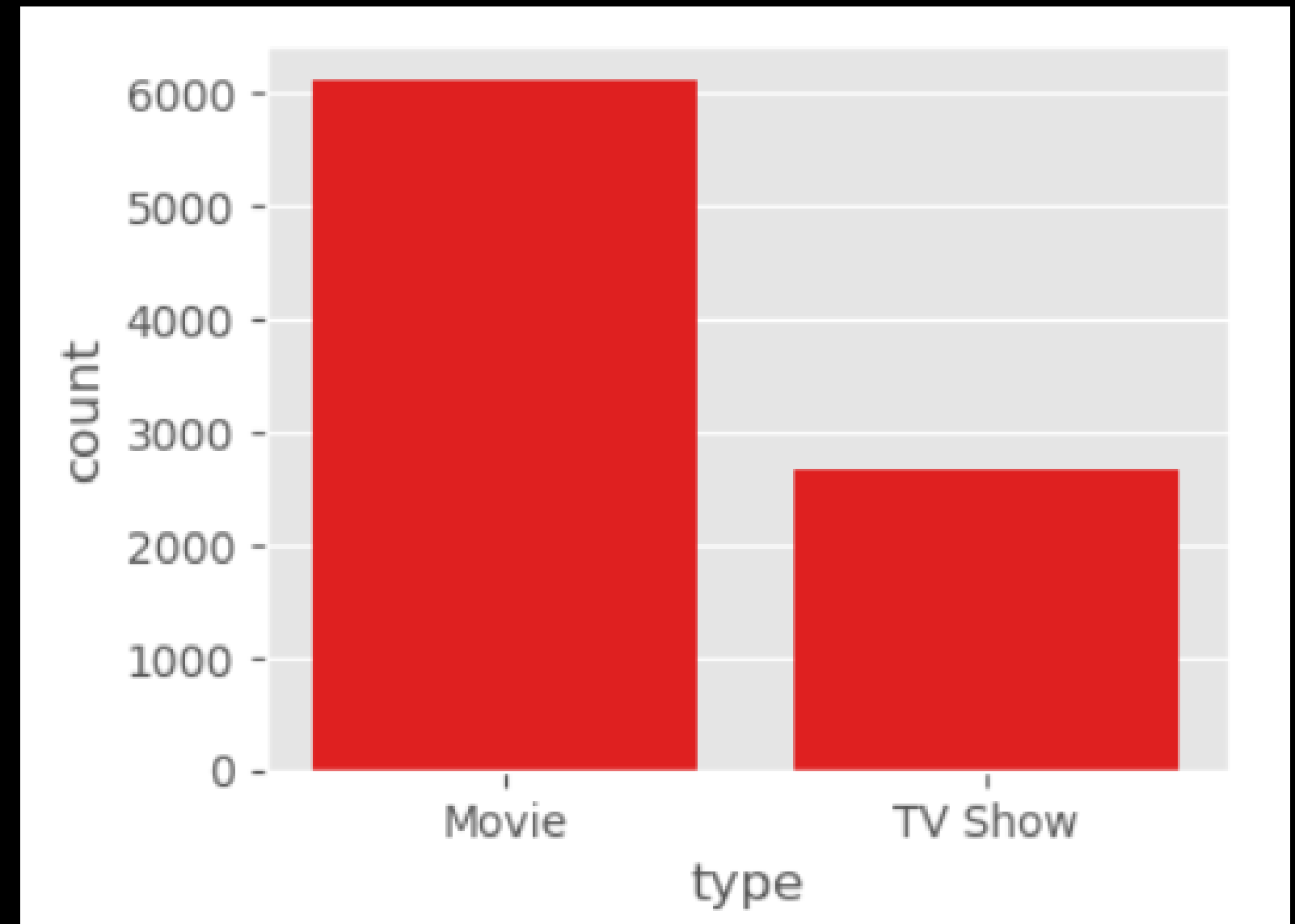
Movies vs TV Shows Analysis

Observation

Movies significantly outnumber TV Shows on Netflix.

Insight

Netflix's content library is primarily movie-focused, providing users with a larger variety of films than series.



VISUALIZATION

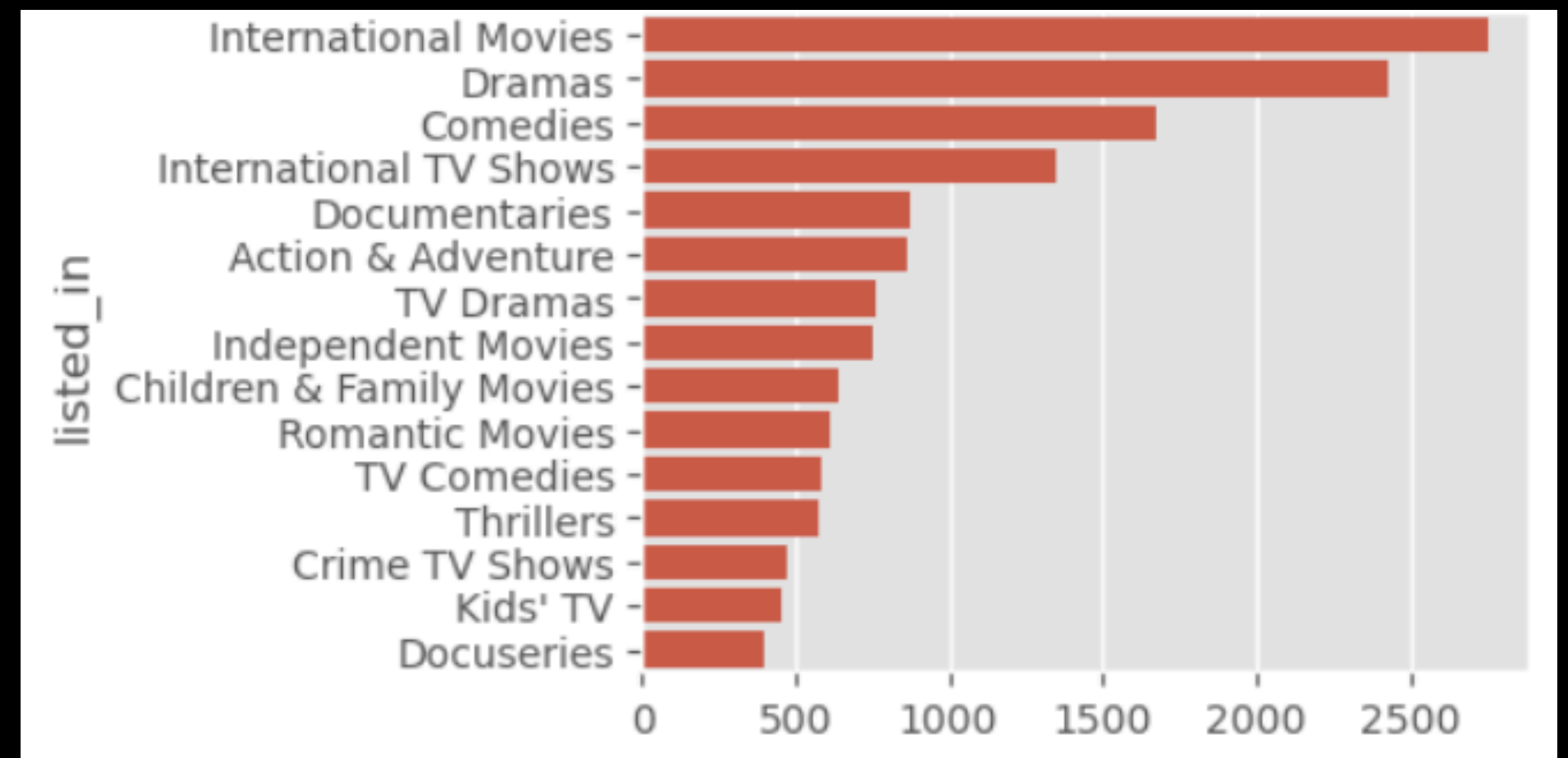
Genre Distribution Analysis

Observation

Drama, International Movies, and Comedy are among the most common genres.

Insight

Netflix prioritizes globally appealing genres that attract a diverse audience across different regions.



VISUALIZATION

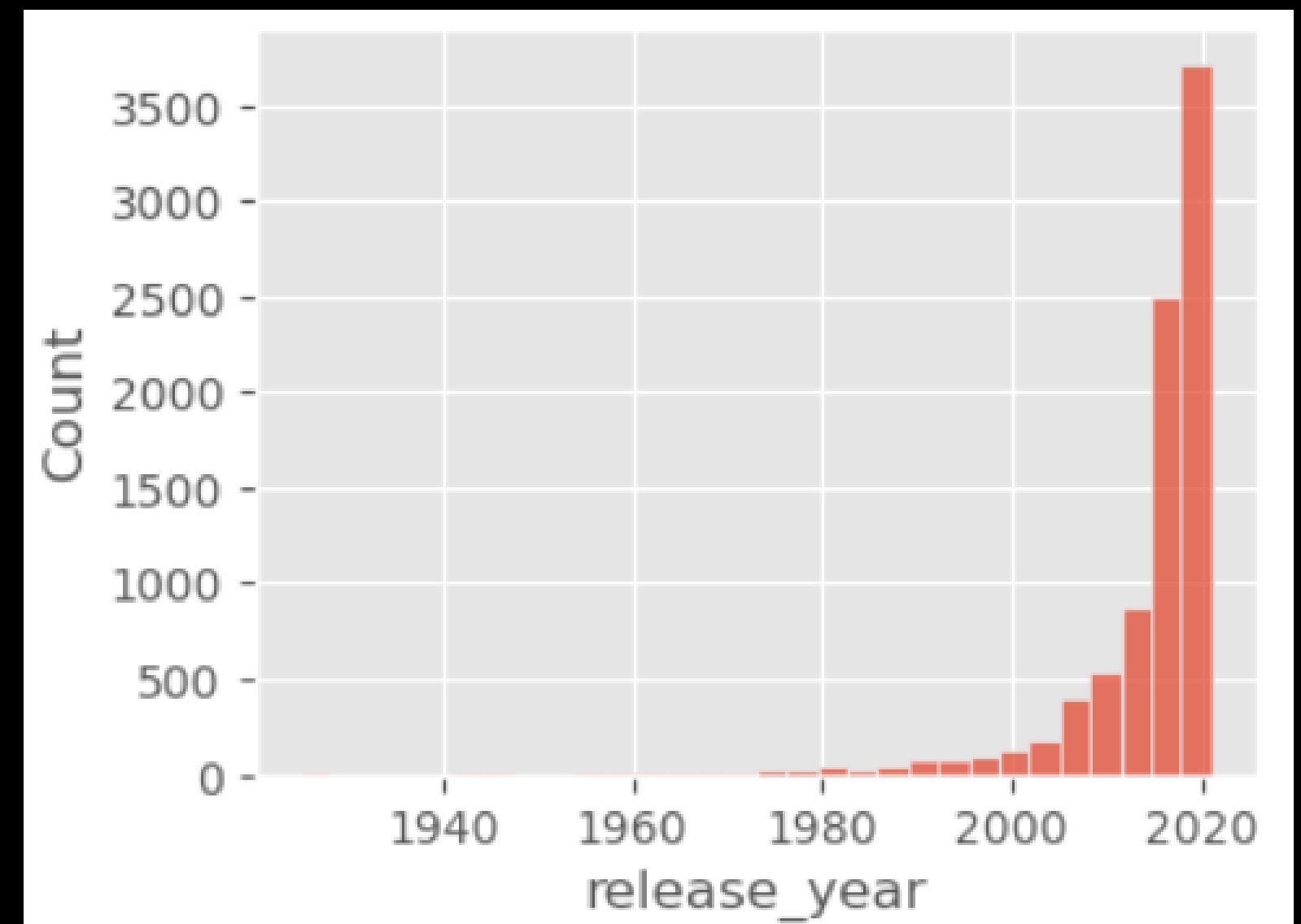
Release Year Distribution

Observation

Most content was released after 2000, with a noticeable increase after 2010.

Insight

Netflix emphasizes modern and recently produced content to match current audience preferences.



VISUALIZATION

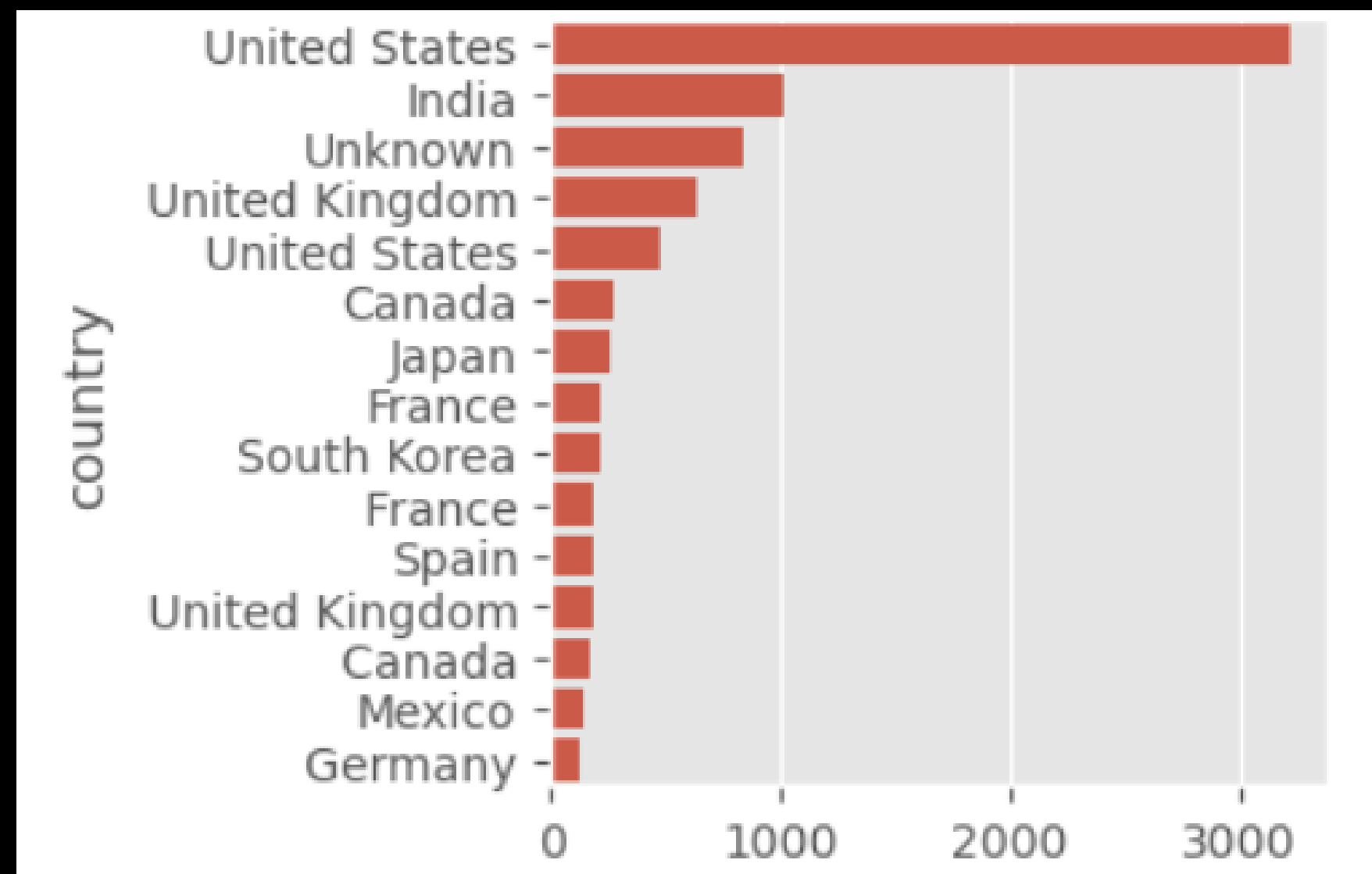
Geographical Distribution Analysis

Observation

The United States contributes the highest number of titles, followed by India and other countries.

Insight

Netflix relies heavily on content from major entertainment industries while expanding internationally.



VISUALIZATION

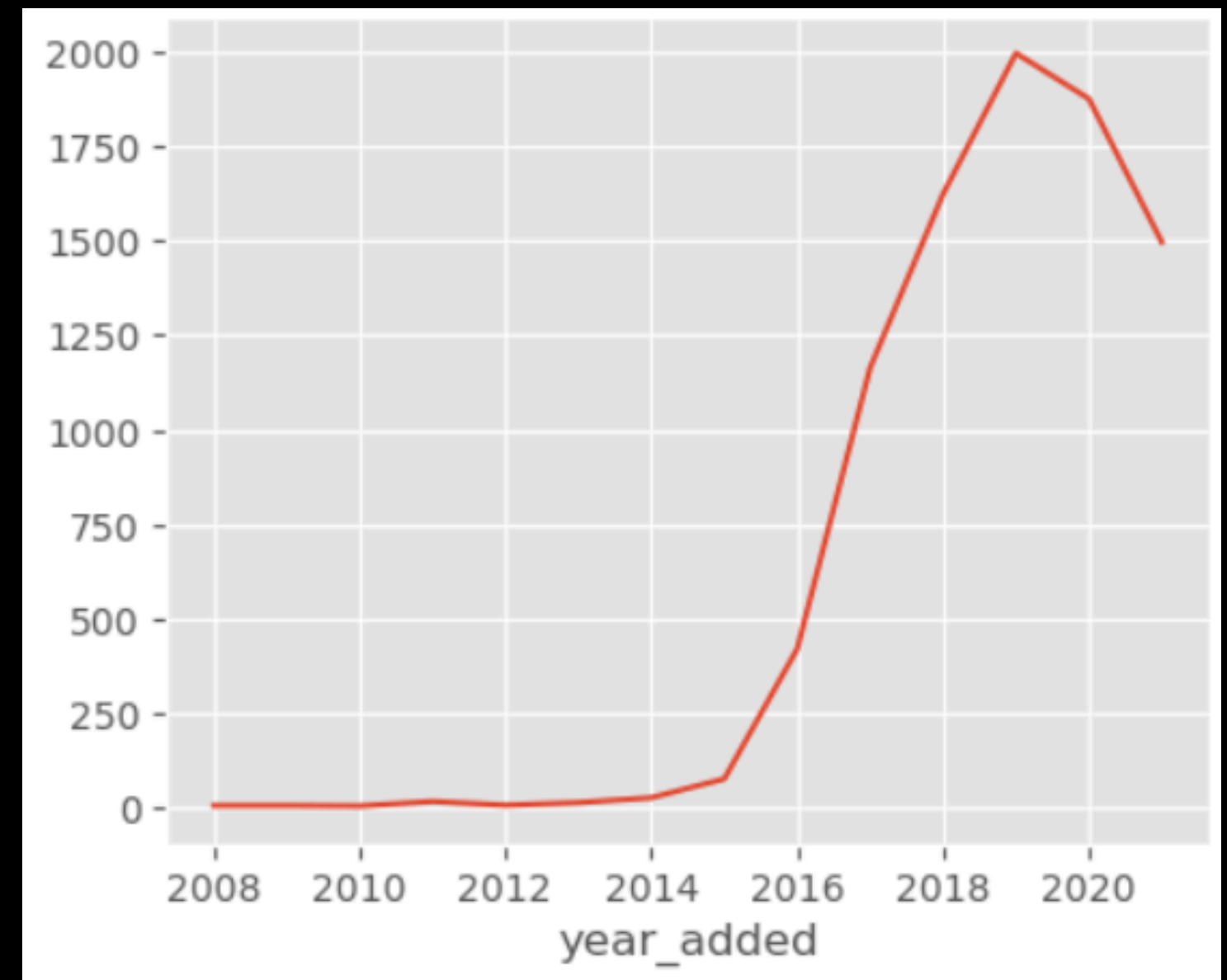
Time Series Analysis

Observation

The number of titles added increased rapidly between 2015 and 2020.

Insight

This period represents Netflix's aggressive content expansion strategy and global market growth.



VISUALIZATION

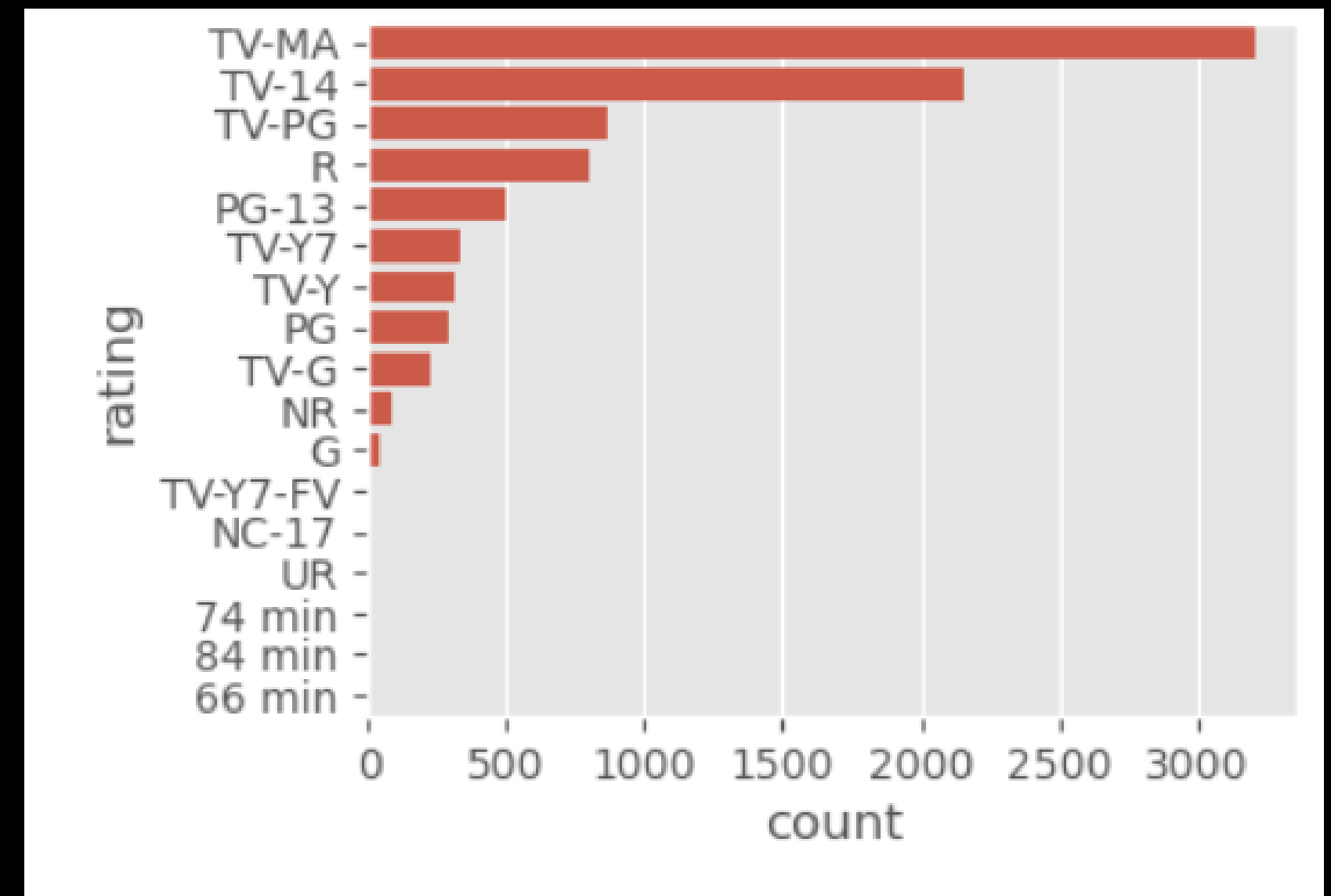
Content Ratings Analysis

Observation

TV-MA and TV-14 are the most common content ratings.

Insight

Netflix mainly targets mature and young-adult audiences through its content offerings.



VISUALIZATION

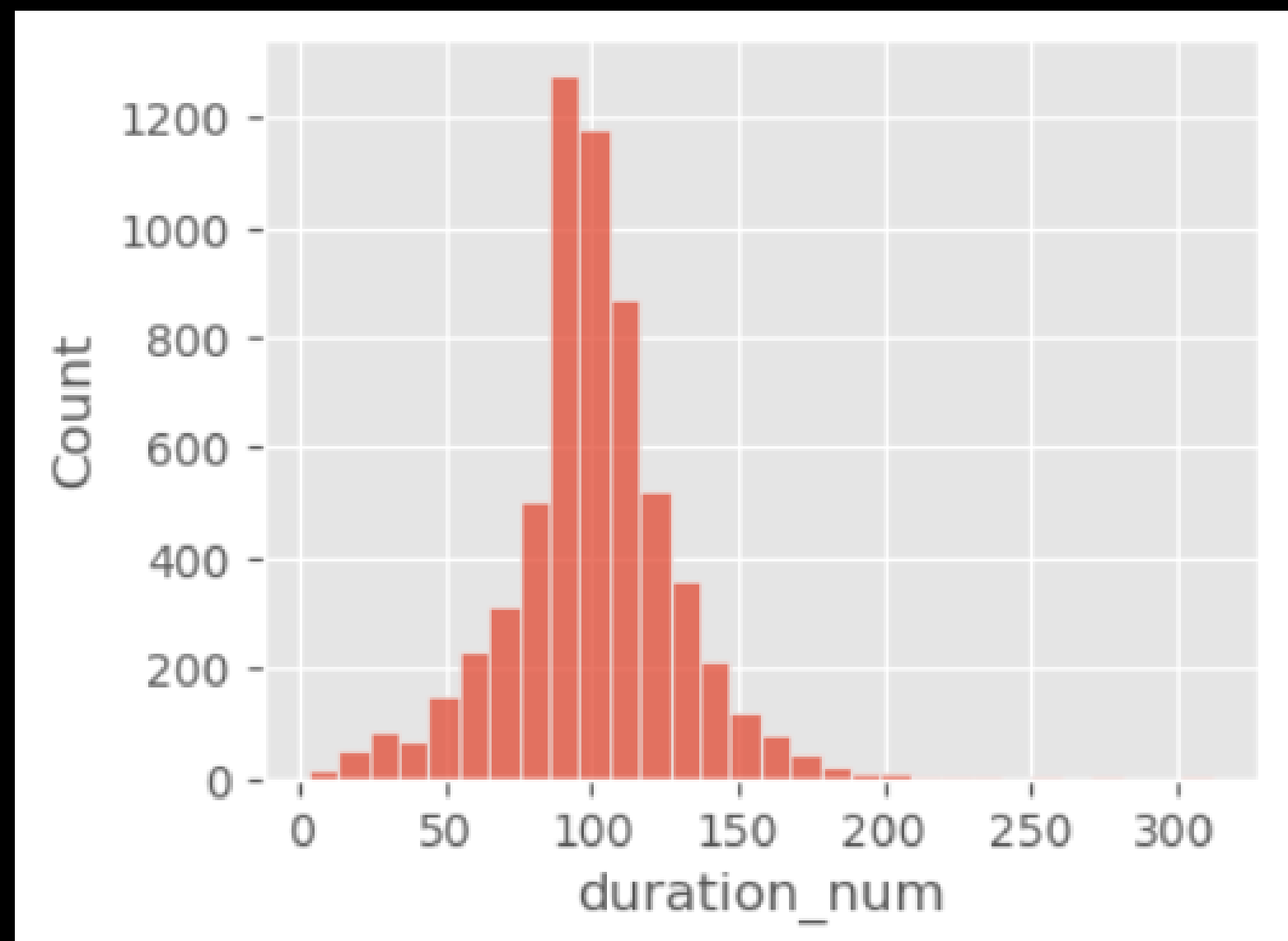
Duration Analysis

Observation

Most movies have durations between 80 and 120 minutes.

Insight

Netflix favors standard-length movies that align with audience viewing preferences.



VISUALIZATION

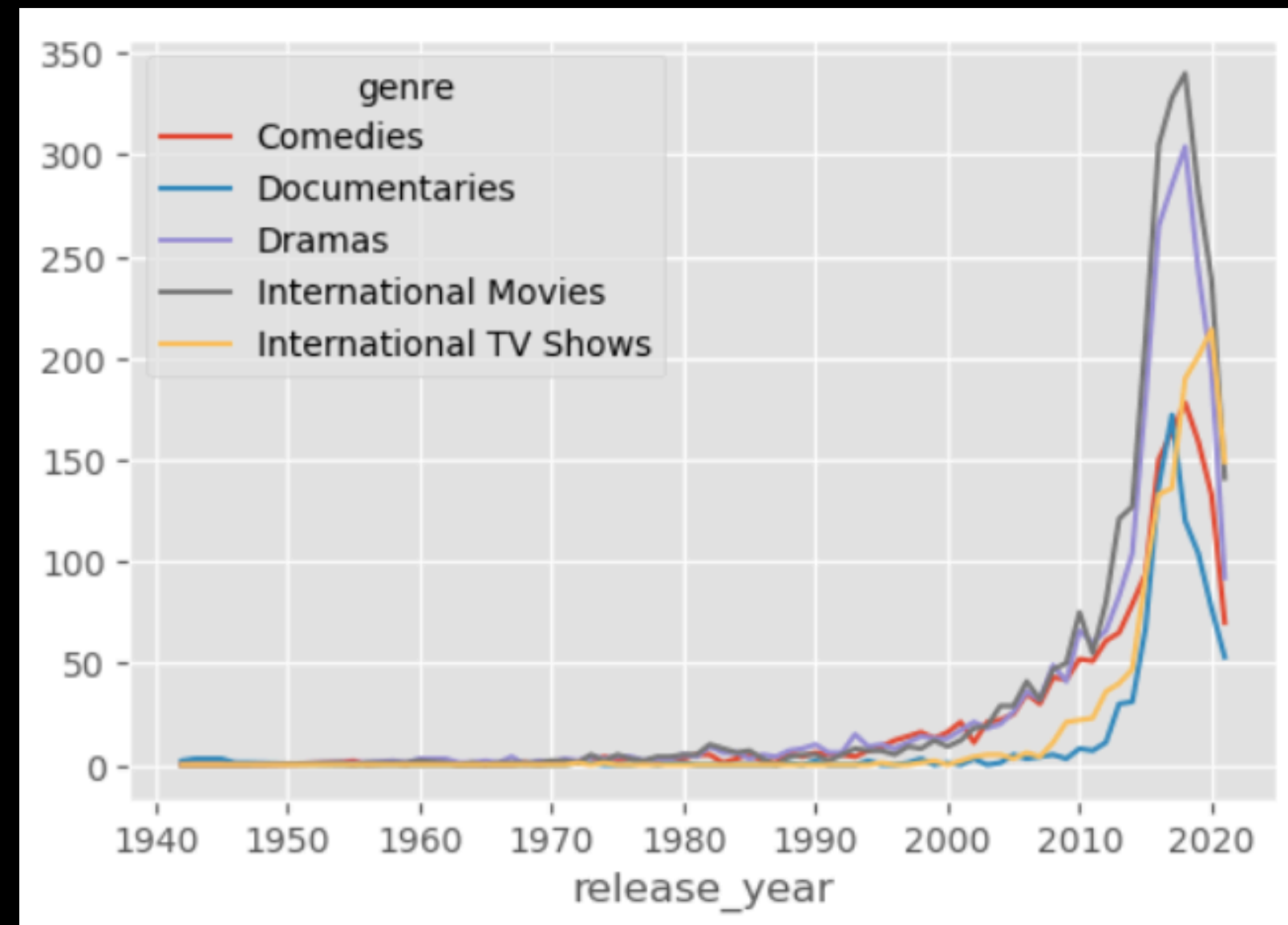
Genre Trends Over Time

Observation

Drama and International content have shown consistent growth over the years.

Insight

Netflix has increasingly invested in diverse global content to attract international subscribers.



VISUALIZATION

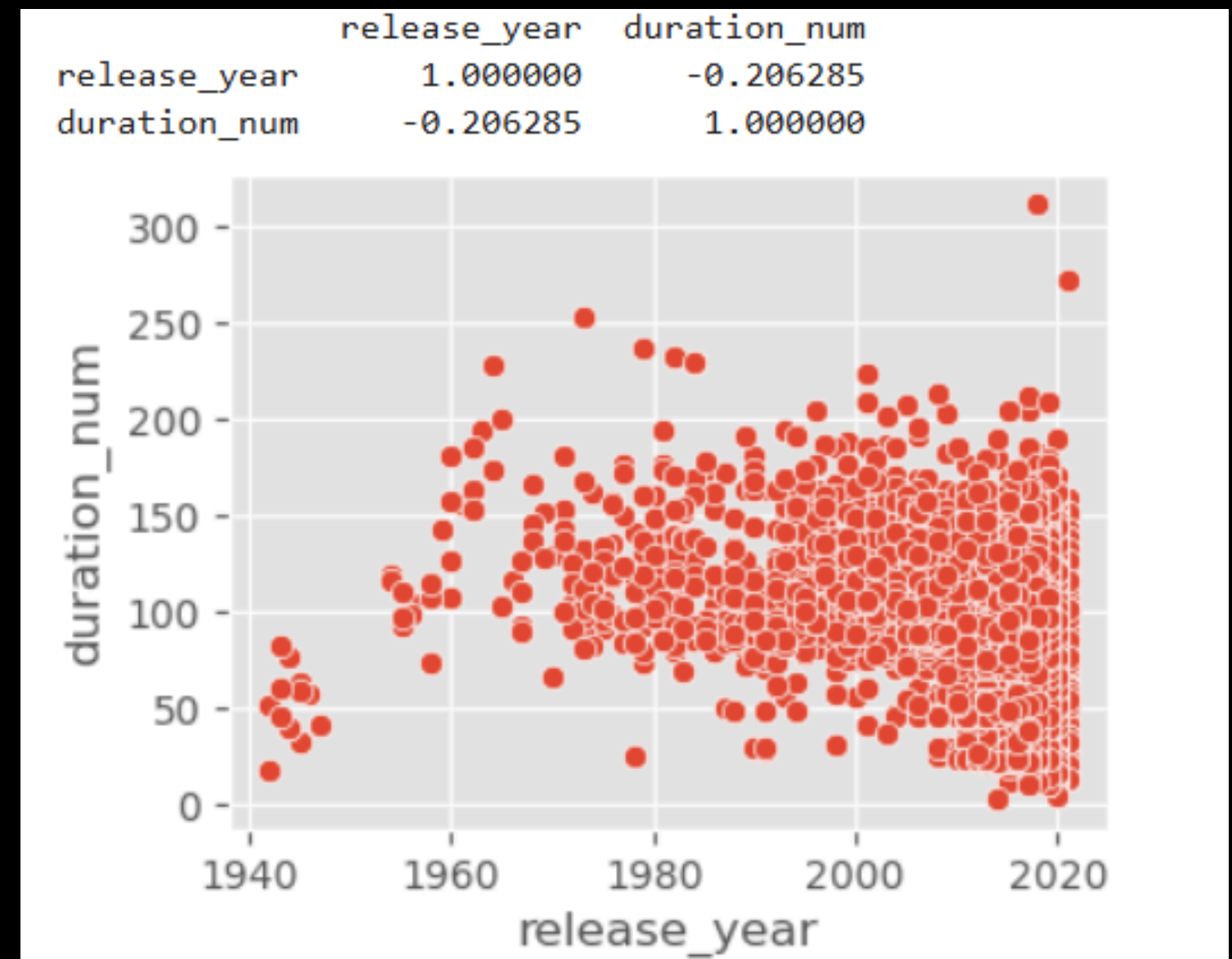
Correlation Analysis

Observation

No strong relationship exists between release year and movie duration.

Insight

Movie length has remained relatively stable despite changes in content production over time.



CONCLUSION



The analysis revealed that Netflix's content catalog is dominated by movies, with Drama and International genres being the most common. Content production and acquisition increased rapidly after 2015, reflecting Netflix's global expansion strategy. The United States remains the largest contributor of content, while other countries are increasingly represented.

RECOMMENDATIONS

- Continue investing in high-performing genres such as Drama and International Movies.
- Increase regional content production to strengthen global reach.
- Maintain a balance between Movies and TV Shows.
- Use historical content trends to guide future content acquisition decisions.
- Expand content diversity to attract wider audience segments.



THANK YOU

FOR CHECKING THIS OUT

GITHUB: https://github.com/mynkbhardwaj87007-spec/netflix_analysis

